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A review of the practical application of micro-aeration and oxygenation for hydrogen sulfide management in anaerobic digesters

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ABSTRACT

With the increased use of biogas generated from anaerobic digestion (AD) as a source of renewable energy, economical methods of cleaning and upgrading the gas are of growing importance. The removal of hydrogen sulfide to varying degrees is required regardless of the end biogas use, as the presence of hydrogen sulfide contributes to operations and maintenance costs and reduction in the life of gas handling equipment. Micro-aeration, or the injection of small amounts of oxygen, into anaerobic digesters has been shown to affect reduction in gaseous hydrogen sulfide concentrations and is more economical than typical sulfide scrubbing systems. This review examines the existing data on micro-aeration applications, including biological kinetics and appropriate control strategies for full-scale micro-aeration applications. Knowledge gaps in research are also noted.

1. Introduction

As methane reduction and sustainable energy production gain prominence in the context of climate and geopolitical change, wastewater and agricultural entities are increasingly looking to beneficially utilize the methane fraction of biogas in renewable energy applications: electrical power, vehicle fuel, renewable natural gas, heating, or in other thermal processes like sludge dryers, gasifiers, or pyrolysis units. Many of these applications require a specified level of treatment to remove biogas impurities which would compromise the end-use equipment or product gas. Among the more problematic constituents in biogas is hydrogen sulfide (H_2S), which, in raw municipal digester gas, averages 50–5000 parts per million (ppm) but can reach into the tens of thousands of ppm (Dumont, 2015) and create corrosive condensates that reduce operational life of a digester gas handling system as well as damage gas pipelines and engines that use biogas as a fuel source (Metcalf and Eddy, 2014).

Historically, adsorptive media, such as granular activated carbon, or wet scrubbing systems using water or chemicals, have been used to remove sulfides from digester gas. These systems require replacement and disposal of spent media and require large amounts of water or chemical disposal, making operations and maintenance more complex. Micro-aeration, or the introduction of small quantities of air or oxygen into anaerobic digesters, is a promising technology as a means of simple, economical, robust, and reliable hydrogen sulfide control (Fdz.-Polanco et al., 2009; Jenicek et al., 2017; Krayzelova et al., 2014; Nguyen and Khanal, 2018; Ramos et al., 2013). The process is primarily biological: sulfide-oxidizing bacteria in the digester convert hydrogen sulfide, in the presence of oxygen, to one of three products, sulfur precipitate, sulfate, or thiosulfate, with all three reactions thermodynamically favorable (Krayzelova et al., 2014).

In addition to the widely known advantages of microaeration i.e. reduction of hydrogen sulfide, improvement of process stability, enhancement of hydrolysis and methane yields (Nguyen and Khanal, 2018; Chen et al., 2020), recently microaeration has been reported to mitigate accumulation of VFA by increasing VFA uptake of facultative hydrolytic bacteria (Guo et al., 2021), improvement of hydrolysis of lignocellulosic substrates (Zeb et al., 2022; Loughrin et al., 2020; Song et al., 2021), improvement of acidification (Valero et al., 2020), hydrogen fermentation (Sołowski, 2022)and enhancing the removal of

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long chain fatty acids (Duarte et al., 2018), organic micropollutants (do Nascimento et al., 2021), and polymeric siloxanes (Ortiz-Ardila et al., 2021).

Compared to other historical means of sulfide removal from digester gas (iron sponge, scrubbers, etc.) microaeration is less expensive, less mechanically complex, and uses little energy relative to the potential sulfide removal (Huertas et al., 2020). Depending on the levels of sulfide required in the end use of the gas, an additional more traditional sulfide removal technology may be required, but the life of the more traditional technology may be enhanced. For example, media replacement can occur less frequently if biogas is pretreated, in effect, through the microaeration process. In the context of biogas as energy, reduction in complexity and operating costs benefits the further adoption of anaerobic digestion as a mature solution to several climate challenges. However, to date, information on the performance and microbial impacts micro-aeration has on digesters has not been fully developed. Recent reviews of micro-aeration (Botheju and Bakke, 2011, Krayzelova et al., 2015, Chen et al., 2020) have focused on the fundamentals of micro-aeration and its beneficial effects on sulfide removal, hydrolysis, and microbial communities. One of the earliest reviews on the subject was the work of Botheju and Bakke which provided a critical review of the effects of oxygen in anaerobic systems, with a focus on various biochemical and physicochemical reactions involving oxygen, without presentation of specific kinetics (Botheju and Bakke, 2011). Krayzelova et al. provided a thorough review of various sulfur oxidizing bacteria, different types of reactors where microaeration was applied such as continuously-mixed tank reactors (CSTR), upflow anaerobic sludge blanket reactors (UASB), expanded granular sludge bed (EGSB), upflow anaerobic filters, and fluidized bed bioreactors, as well as Monod kinetics for four pure SOB cultures (Krayzelova et al., 2015). Koshnevisan et al. reviewed biological H_2S removal from biogas, with a particular emphasis on biotrickling filters and microaeration, and highlighted the importance of optimizing headspace design for microaeration, as well as specifying a minimum biogas residence time of 5 h for successful biogas desulfurization in anaerobic digesters (Khoshnevisan et al., 2017). Chen et al. reviewed the applications of microaeration in anaerobic digesters focusing on process control and impact on various metabolic pathways (Chen et al., 2020). Nguyen and Khanal (2018) provided an extensive review of the bioenergetics of microaeration, elaborating on the microbial metabolic pathways, and the syntrophic relationships between various microbial groups in microaerobic environments. The aforementioned study presented the kinetics of various microbial groups, with only one set for sulfide oxidizing bacteria (SOB).

Despite the critical role played by SOB in biogas desulfurization, data on the kinetics of SOB (associated with sulfide removal from biogas streams) have been collected inconsistently and largely from bench-scale studies that employ test conditions not reflective of actual operating parameters. Additionally, more can be done to assess physical parameters of digester geometry and design to evaluate potential removal efficiencies for existing digesters. For example, while headspace in digesters has been identified as an important parameter in evaluating sulfide removal efficiency (Ramos et al., 2012, 2014a), other digester design and construction elements such as the liquid surface area-to-volume ratio have been overlooked.

This study, which focuses on the application of microaeration to reduce H_2S in municipal wastewater treatment plant digesters, begins to fill the identified knowledge gaps i.e. absence of reliable full-scale empirical design criteria, dispersed SOB kinetics that hinder process modeling, and sparse information on instrumentation and its limitations, by critically reviewing full-scale applications and, to the authors' knowledge, develop for the first time correlations between microaeration air requirements and the amount of sulfur oxidized. Additionally, a first-order rate constant for desulfurization in digester headspaces has been derived that can be used to ascertain the removal efficiency of micro-aeration in existing digester infrastructure. The correlations and rate constant can be applied by practitioners and utilities in microaeration system design to evaluate sulfide removal efficiency in the context of other more costly sulfur removal systems. Moreover, considering the limited modeling studies on microaeration (Donoso-Bravo et al., 2018; Pokorna-Krayzelova et al., 2017a) we collected and presented comprehensive Monod-kinetic parameters for sulfur oxidizing bacteria and reviewed operational and kinetic parameters of micro-aeration systems while identifying knowledge gaps in each area. The study also expanded the review of online instrumentation, highlighting the limitations of current analytical methods, and their impact on automated process control.

2. Hydrogen sulfide generation in digesters

Hydrogen sulfide is generated in anaerobic digesters primarily through the decomposition of organic matter and the use of sulfate as a terminal electron acceptor by sulfate-reducing bacteria (SRB). The contribution of sulfate reduction to hydrogen sulfide production is variable; Ramos et al. reported that variability in sulfate can be significant, with carbon and sulfur contents ranging from 29% to 37% and 0.5–2.0% w/w, respectively (Ramos and Fdz-Polanco, 2014).

Hydrogen sulfide concentrations in the biogas are dictated primarily by the sulfur content of the feed and its methane yield. Peu et al. (2012) evaluated the hydrogen sulfide content of various feedstocks including wastewater treatment biosolids, crop residues, animal wastes, agri-food industrial wastewater treatment facilities, grease traps, and various components of municipal solid wastes, and reported that H_2S concentrations from biological sludges at wastewater treatment plants can reach as high as 19,000 ppm. On the other hand, fats and grease were at the other end of the spectrum, with H_2S concentrations of less than 1000 ppm.

A significant source of sulfur in anaerobic digesters comes from the hydrolysis and fermentation of proteins and associated amino acids, resulting in a release of both hydrogen sulfide and methylated sulfurs (Higgins et al., 2008). Higgins et al. demonstrated that methylated sulfur compounds (ex. methanethiol) are converted to methane through methylotrophic methanogenesis, indicating that organic bound sulfur is a significant contribution to digester gas hydrogen sulfide levels. A later study by Higgins demonstrated that during the decomposition of dewatered sewage sludge, a static headspace could accumulate organic sulfurs (methanethiol, etc.) and hydrogen sulfide, a process attributable primarily to the decomposition of methionine and cystine (Higgins et al., 2008). The decomposition of these constituents, in turn, results in the release of methyl mercaptan (converted to methionine), which is further reduced to hydrogen sulfide, and hydrogen sulfide directly (converted to cystine). Additionally, Du and Parker found that both sludge type and digestion conditions impact the bioavailable sulfur content: under mesophilic digestion, bioavailable sulfur comprises 32% of the total influent sulfur for municipal primary sludge and 61% for waste activated sludge, increasing under thermophilic conditions to 46% and 72%, respectively (Du and Parker, 2013), which will also impact H_2S production in the digester.

The impact of SRB on anaerobic processes has been reported to extend beyond the generation of hydrogen sulfide to reductions in methane production through the consumption of organic matter. In work by Uberoi and Bhattacharya, it was noted that that the chemical oxygen demand (COD) to sulfate (SO_4^{2-}) ratio influenced the magnitude of the SRB's influence on anaerobic digestion as metabolism decreased with increasing COD load. Also noted was that volatile acid selectivity by SRBs was dependent upon the availability of SO_4^{2-}; at ratios less than 0.44 of propionate to SO_4^{2-}, acetate utilization increased (Uberoi and Bhattacharya, 1995).

3. Chemical oxidation of hydrogen sulfide in anaerobic digesters

Chemical H_2S oxidation models are summarized in Table 1. The basic relation for the kinetics model was expressed in O'Brien and Birkner

Table 1
Chemical Oxidation Model (adopted from Krayzelova et al., 2015).

Chemical H_2S oxidation	Description	Reference
$k_m \cdot (S_{H2S})^{\alpha} \cdot (S_{O2})^{\beta}$	km is the rate constant (min^{-1}), S_{H2S} is the H_2S concentration (mmoL/L), S_{O2} is the O_2 concentration (mmoL/L), α is the reaction order with respect to the sulfide concentration (–), β is the reaction order with respect to the oxygen concentration (–).	(O'Brien and Birkner, 1977)
	k (min^{-1}) α β S_{H2S} (mmolS^{2-}/L) S_{O2} (mmolO_2/L)	
	17.46 1.02 0.8 0–5 0.15	(Klok et al., 2013)
	0.1165 1.00 1 0.04–0.1 Saturated (25 °C)	(Luther et al., 2011)
	0.57 0.41 0.39 0.16–9.38 0.003–0.266	(Buisman et al., 1990)
	0.055 0.38 0.21 0.09–0.3 0.16–0.62	(Wilmot et al., 1988)
	67.6 1.15 0.69 0.05–0.2 0.6	(Jolley and Forster, 1985)
	1.44 1.02 0.8 0.02–1.21 0.21–1.1	(O'Brien and Birkner, 1977)
$\frac{k_0 + k_1 \cdot K_1 / S_{H+}}{1 + K_1 / S_{H+}} \cdot (S_{S^{2-}})^{\alpha} \cdot (S_{O2})^{\beta} \cdot \theta^{T-20}$	S_S^{2-} is the concentration of total sulfide (mg/L), k_0 and k_1 are the rate constants for the oxidation of H_2S and HS^-, respectively [(mgS/L)$^{1-\alpha}$ (mg O_2/L)$^{-\beta}$ /h], θ is the Arrhenius constant = 1.06, T is the temperature (°C) = 5–25 °C, K_1 is the first dissociation constant for H_2S (approx.1.0 $\times 10^{-7}$). The reaction order α and β were 0.9 and 0.2 respectively, θ was 1.06, and k_0 and k_1 fluctuated from 0.02 to 0.08 and from 0.25 to 1.00, respectively. Based on molar concentrations and the α and β values, k_0 * = 0.022k_0, and k_1 * = 0.03125 k_1	(Nielsen et al., 2006)
$k_m \cdot (S_{H2S})^{\alpha} \frac{S_{O2}}{K_{S,O2} + S_{O2}}$	k_m 4.46 h^{-1}, α 0.56, and K_{O2} 1.30 mg/L. H_2S oxidation rate was independent of the O_2 concentration at the O_2 concentration above 5 mg/L (0.156 mM). If H_2S concentrations in mmole instead of mg/L, the rate is to be multiplied by 0.1436	(Sharma et al., 2014)

(O'Brien and Birkner, 1977) using the terms of the rate constant, H_2S and O_2 concentrations, and reaction orders (α and β). Various studies used this model and determined rate constants (k) ranging between 0.1 and 67 min^{-1}, α values of 0.38–1.15, and β values of 0.21–1.0. The wide variation was related to different study conditions, including sulfide and oxygen concentrations (0–9.38 mmol and 0.003–1.1 mmol, respectively), analytical methods, and buffer solution. Analyzing the effects of pH and temperature on the stoichiometry of chemical oxidation, Nielsen

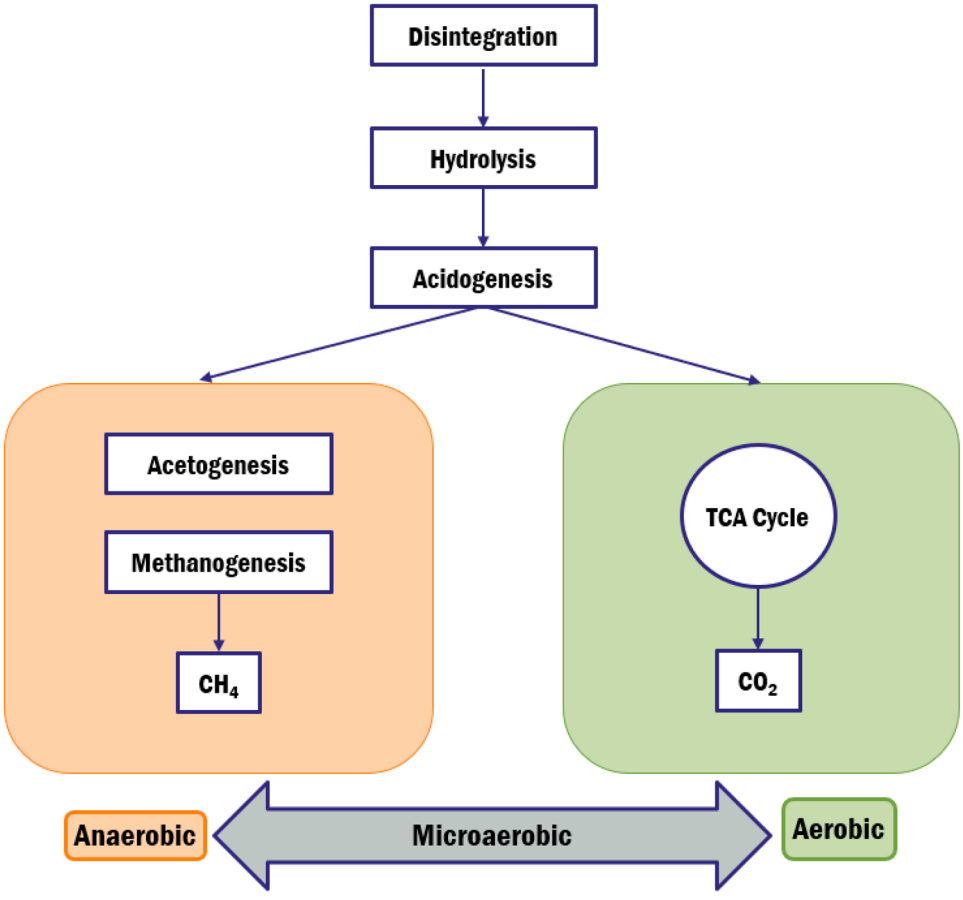


Fig. 1. Co-function of Anaerobic and Aerobic Metabolisms in Anaerobic Digestion Adapted from (Chen et al., 2020).

et al. found that the rate of chemical sulfide oxidation could be related to the dissociation of H_2S to HS^-, with HS^- being oxidized at a higher rate than H_2S stoichiometry of chemical oxidation was not significantly affected by varying pH (6−9) and temperature (5–25 °C) (Nielsen et al., 2006).

4. Application of micro-aeration and its effects on anaerobic digestion

Fig. 1 illustrates the general pathways for degradation of organic matter in anaerobic digesters in the presence and absence of microaeration. Pyruvate, the key product of acidogenic bacteria, is broken anaerobically to acetylCoA, which undergoes transformations to hydrogen, carbon dioxide, formate, acetate, ethanol, lactate, propionate, and butyrate, and then converted to methane via hdyrogentrophic and acetoclastic methanogens. Under aerobic conditions, acetylCoA is converted to carbon dioxide. A review by Nguyen and Khanal indicated that micro-aeration contributed to enhancements in the rate of hydrolysis, typically the rate limiting step of digestion (Nguyen and Khanal, 2018). Stabilization of volatile fatty acid accumulation in the digesters was also noted. The transformation of lactic, valeric, butyric, and propionic acids to acetate has also been enhanced in the presence of oxygen (Lim and Wang, 2013).

The impact of micro-aeration on volatile solids (VS) reduction and methane yields, however, is controversial. Jenicek et al. noted decreases in the overall sludge volatile fraction with micro-aeration (1.6–9.3%) and average increases in biogas and methane production of 17% and 16%, respectively, suggesting enhanced overall biodegradation due to the combined micro-aerobic and anaerobic metabolisms (Jenicek et al., 2017). Fig. 1.

On the other hand, Diak et al. observed that although micro-aeration of municipal primary sludge enhanced the generation of soluble proteins and ammonia, it had no effect on the degradation of carbohydrates and the reduction of VS (Diak et al., 2013). Further efforts on full scale application of micro-aeration and its effects on AD as a whole are warranted.

5. Noted impacts on the microbial species and interactions in micro-aerobic environments

In addition to the aforementioned studies on hydrolysis, acetogenesis, etc., several studies have focused on the effects micro-aeration has on the microbiome and microbial interactions in anaerobic digesters. Ruan et al. investigated the impacts of micro-aeration on hydrolysis rates and methane production. The impact of micro-aeration was largely evident in the bacterial populations but not archaea populations primarily associated with methane production (Ruan et al., 2019). Table 2 summarizes the changes at the phyla and genus levels reported by the aforementioned study.

Tang et al. observed a shift from a predominance of *Methanosarcina* sp. (an archaea) to *Methanoculleus* sp. (a bacteria) with micro-aeration because of the latter's higher oxygen tolerance (Tang et al., 2004). Conversely, Ramos et al. noted minimal changes on archaeal populations between reactors with and without micro-aeration but noted significant differences in bacterial populations (Ramos et al., 2014b).

Song et al. evaluated desulfurization through air introduction into the headspace of an anaerobic digester treating chicken manure. The study not only reduced H_2S in the digester from 5000–6000 ppm to 10–15 ppm and improved methane yield by 6%, but it also reported an increase in the relative abundance of hydrogenotrophic methanogens such as *Methanobacterium*. The enrichment of syntrophic bacteria (*Syntrophomonas*) accelerated a synergistic interaction between hydrogen-producing bacteria and hydrogenotrophic methanogens. This in turn resulted in fewer H_2-utilizing SRB, which compete with methanogens. (Song et al., 2020).

Table 2
Summary of Genus and Phyla in Microaerobic Systems (Ruan et al., 2019).

Phylum	Primary Function	Percent Change in Relative Abundance Micro-Aeration vs. Conventional Anaerobic Digestion
Proteobacteria	Hydrolysis and fermentation and induced cell lysis	+ 15.3
Chloroflexi	Soluble microbial products (SMP) degradation through hydrolysis	+ 37.9
Synergistetes	Volatile fatty acids to Acetate	+ 77.7
Actinobacteria	SMP degradation through hydrolysis	+ 58.3
Genus		
Thermovirga	Amino acid metabolism through elemental sulfur reduction	+ 117%
Norank_Anaerolineaceae	Chemoheterotrophs that utilize sugars and proteinaceous compounds	+ 49.9
Norank_Caldilineaceae	Chemoheterotrophs that utilize sugars and proteinaceous compounds	+ 34.4
Candidatus_Microthrix	Lipid metabolism through long chain fatty acids	+ 92

6. Kinetics of H_2S bio-oxidation

The kinetics of sulfate oxidizing bacteria (SOB) were estimated in several studies with various operational conditions as summarized in Table 3. HRTs of completely stirred tank reactor (CSTR) systems varied from 6 to 8 hrs to 20 days while volumetric sulfur loading rates ranged from 0.007 to 24 g/L/d (Donoso-Bravo et al., 2018). While most studies operated at one temperature, only two studies tested different temperatures: Nielsen et al. evaluated reactors at both 10° and 20 °C (Nielsen et al., 2006) and Wilmot et al. at a broader range of 10° and 40 °C (Wilmot et al., 1988). Most studies were conducted at a pH of 7–8 and DO concentrations were more than 3 mg/L except for one study that operated at a DO of < 0.1 mg/L (Pokorna-Krayzelova et al., 2018). Additionally, most studies used pure cultures of SOB, though activated sludge (Munz et al., 2009; Nielsen and Vollertsen, 2021), anaerobic digestate (Donoso-Bravo et al., 2018), and municipal wastewater biomass (Nielsen et al., 2006; Wilmot et al., 1988) were also employed.

Despite the inconsistent biomass yield units, the values were lower than 0.22 mgVSS/mgS or 0.11 mgVSS/mgCOD using a COD equivalent of sulfide of 2 (8e-eq of O_2 per moleS*(8gO2/e- eq)/32 gS/mole), which was lower than the heterotrophs of 0.4 mgVSS/mgCOD (Metcalf and Eddy, 2014). Moreover, reported yields showed a wide variation due to the different types of SOBs. For example, the yield of SOB oxidizing sulfide to sulfur were 0.0125 and 0.038 gCOD/gS while the yield of SOB oxidizing sulfide to sulfur and then to sulfate was 0.175 and 0.15 gCOD/gS (Buisman et al., 1990; Munz et al., 2009), indicating that the yield of SOB responsible for oxidizing sulfide to sulfur was much lower than that of SOB oxidizing sulfur to sulfate.

Sulfide affinity coefficients (Ks, S^{2-}) also varied widely between < 0.1 mgS^{2-}/L (Pokorna-Krayzelova et al., 2018) to 11 mgS^{2-}/L (Xu et al., 2013b). One study reported an oxygen affinity coefficient of 200 mg/L; however, this was obtained by supplying pure oxygen to the headspace at 250 mg/L to 1500 mg/L to treat 200 mgS^{2-}/L (Xu et al., 2013).

The maximum specific growth rates (μSOB) were estimated as < 1–7 d^{-1} except for the unrealistically high 40 d^{-1} in Donoso-Bravo et al. (Donoso-Bravo et al., 2018), which is higher than typical growth rates of autotropic bacteria (< 1 d^{-1}) (Metcalf, and Eddy, 2014). Decay rates of SOB were 0.01–0.13 d^{-1}, except for 5.76 d^{-1} (Pokorna-Krayzelova et al., 2018). The maximum substrate utilization rates ($k_{m,SOB}$) were 10.4–83

Table 3
Biochemical Oxidation Model Parameters and Test Conditions.

km_{SOB}	b_{SOB} (d^{-1})	μ_{SOB} (d^{-1})	$K_{s,\ s2-}$ (mgS^{2-}/L)	$K_{s,\ O2}$ (mg/L)	Y_{SOB} [(c)]	DO (mg/L)	Sulfide (mg/L) [load g/L/d]	Temp (°C)	pH	Experimental setup	Dominant organisms	Ref.
7.4 mgS^{2-}/mgVSS/d or 10.4/d	N/A	0.67	11	200	0.09 mgVSS/mgS^{2-}	See note[(f)]	200	23	8	250 mL batch tests	SOB from activated sludge	(Xu et al., 2013)
N/A	0.13	N/A	N/A	N/A	See note[(d)]	4	500 [2]	20	8	MBR (HRT 6 hr & SRT 5d) Respirometric tests	SOB of *r-Proteobacteria* and *Halothiobacillaceae* class	(Munz et al., 2009)
32.3 mgHS/$mg_{protein}$/d	N/A	2.8	8.96	N/A	0.0891 $mg_{protein}/mgS^{2-}$	See note[(g)]	System (4000, [load 0.3]) Resp. (0–256)	30	7	CSTR (HRT 13 d), Respirometric tests	*Thiobacilli* sp.	(Alcántara et al., 2004)
74 mgS^{2-}/$mg_{protein}$/d	N/A	7.2	0.32	N/A	0.0969 $mg_{protein}/mgS^{2-}$	N/A	93–304	27	7.6	Continuous culture	Pure culture of *Thiobacillus thioparus*	(de Zwart et al., 1997)
160 d^{-1} [(l)]	0.02	40		3	0.25 mg/mg	N/A	245 [0.01]	35	7.2	Pilot-scale digester (HRT 20d)	Mesophilic anaerobes	(Donoso-Bravo et al., 2018)
34 d^{-1} [(l)]	N/A	N/A	N/A	N/A	N/A	N/A	N/A	32	7.4	1.5 L Fed-batch bioreactor	*Thiomicrospira* sp. CVO	(McComas et al., 2001)
82.3 d^{-1} [(l)]	0.01	6.6	3.2	3.2	0.08 mgCOD/mgCOD	1–7.4 [(h)]	I 24 [0.07][(j)]	35	7.1	3 L CSTR (HRT 7.9 hr)	NA (granular sludge)	(Pokorna-Krayzelova et al., 2017a)
21.8[(a)] mgS/mgCOD/d or 43.6/d	5.76	7	0.032	3.2	0.32 mgCOD/mgS or 0.22 mgVSS/mgS	< 0.1	8–10	35	7	2 L batch reactor	*Sulfuricurvum kujiense*	(Pokorna-Krayzelova et al., 2018)
21.8[(b)] mgS/mgCOD/d or 43.6/d	0.06	7	0.032	3.2	0.32 mgCOD/mgS or 0.22 mgVSS/mgS	See note[(i)]	165 [0.007]	35	7.5–7.7	13 L SBR[(k)]	*Thiomonas intermedia*	(Andreides et al., 2021)
11.6 mgS/mgCOD/d or 23.2 /d	N/A	1.98	0.1	N/A	0.17 mgCOD/mgS or 0.12 mgVSS/mgS	100% saturation	2.5	20	7.9	Batch OUR tests	Activated sludge	(Nielsen and Vollertsen, 2021)
N/A	N/A	N/A	N/A	N/A	N/A	4–8	10	10, 20	5, 8, 9	3.4 L batch reactor	Wastewater biomass	(Nielsen et al., 2006)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	10, 20 [8.4, 17]	28–30	N/A	Bioreactor (150 L/h, HRT 1.7 min)	*Thiobacillus Thioparus Ch11*	(Chung et al., 1996)
N/A	N/A	N/A	N/A	N/A	See note[(e)]	3–7	[1.5–23.7]	23	N/A	8.3 L reactor (continuous)	N/A	(Buisman et al., 1991)
N/A	N/A	N/A	N/A	3	N/A	5–20 (initial)	0.2–8	10–40	7	2.4 L reactor batch test	Wastewater biomass	(Wilmot et al., 1988)

Notes.
a. Pokorna-Krayzelova et al. (2018) incorrectly reported 482 mmol S/mg COD/h instead of 21.8 mgS/mgCOD/d or 43.6/d.
b. Andreides et al. (2021) adopted data from Pokorna-Krayzelova et al. (2018).
c. Conversion factor of VSS to COD is 1.42
d. 0.038 $mgCOD/mgS^{2-}$ or 0.027 $mgVSS/mgS^{2-}$ for SOB oxidizing sulfide to sulfur, 0.116 $mgCOD/mgS^{2-}$, 0.15 $mgCOD/mgS^{2-}$ for SOB oxidizing sulfide to sulfate
e. 0.0125 mgCOD/mgS or 0.09 mgVSS/mgS for SOB oxidizing sulfide to sulfur, 0.175 mgCOD/mgS for SOB oxidizing sulfur to sulfate
f. Pure oxygen was added to the headspace (50 mg/L to 250 mg/L) to make a molar ratio of O_2/S^{2-} 0–2.5. The ratio of 1 was used for modeling
g. Molar ratios of O_2/S^{2-} 0–2 with an optimum of 0.5 and DO was supplied from oxygen tank of 5 mg/L
h. 1–7.4 mg/L was used for simulations
i. Air was added to the headspace $O_2/\ S^{2-}$ ratio of 1
j. 160–496 mgS/L was used for simulations
k. Cycle: feeding 2.5 min, reaction 210 min, settling 50 min, HRT of 22 days
l. μmax reported

d^{-1}, except for 160 d^{-1} (Donoso-Bravo et al., 2018), which is much higher than heterotrophs (13 d^{-1}) and nitrifiers (6 d^{-1}). Generally, biokinetics derived from pure cultures of SOB were significantly higher than from mixed cultures, with the two studies using SOBs from activated sludge reporting μSOB of 0.67–1.98 d^{-1}, much lower than pure cultures (Nielsen and Vollertsen, 2021; Xu et al., 2013). The wide variability in pure culture kinetics may also be attributable to variability in oxidized sulfur products, as the biomass yields are different as discussed above. The most salient feature of the 14 studies reported in Table 3 is operation at high DOs.

Most mathematical models used Monod equations and oxygen uptake rates (OUR) from respirometry tests to estimate kinetics (Andreides et al., 2021; Munz et al., 2009; de Zwart, at al. 1997; Pokorna-Krayzelova et al., 2018; Pokorna-Krayzelova et al., 2017a). Two studies used modified anaerobic digestion models. Pokorna-Krayzelova et al. (2017a) expanded the ADM1 model by adding five processes, four for SRB activity and one for SOB activity converting hydrogen sulfide to sulfur. The model was calibrated with experimental data from a mesophilic 3 L-bioreactor with 10% headspace, fed with synthetic wastewater (2.32 g COD/L and 0.072 g SO42-/L), and operated at an HRT of 7.9 h for 373 days with micro-aeration (0.001 m^3/d) starting on day 74. Sulfur mass balances for the micro-aerated reactor showed that the simulated H_2S in biogas (6% of total) was similar to the experiment (9%), despite the simulated H_2S concentration in the biogas being significantly lower than experimental data. The higher removal prediction is likely because the simulation did not reflect aerobic carbon oxidation and re-reduction of elemental sulfur back to sulfide(Pokorna-Krayzelova et al., 2017a).

The Andreides et al. study modified Pokorna-Kravelova's model by considering liquid phase processes that affect H_2S concentration in the headspace, then developed a general model of micro-aeration for intermittently mixed and/or fed CSTR or SBR digesters (Andreides et al., 2021). Seven processes influencing the concentration of sulfur forms in the reactor were included in this model: uptake of sulfate by SRB in the liquid, chemical oxidation of H_2S in the headspace, uptake of H_2S by SOB in the biofilm, uptake of S^0 by SOB, SOB decay, SRB decay, and washout of H_2S from the reactor (Table 3). Simulation kinetics were adopted from Pokorna-Kravelova, and the model was validated using experimental data collected from a mesophilic microaerobic reactor (working volume of 13 L and head space 7 L, HRT of 22 d, OLR of 0.9 g/L/d) fed with cheese whey (20 g/L) mixed with K_2SO_4 (0.9 g/L) (496 mg/L as SO_4).

To understand the two pathways of sulfide oxidation via chemical and biochemical reactions, an integrated mathematical model including chemical and biochemical oxidation was also developed by Pokorna-Krayzelova et al., who monitored the chemical and biochemical oxidation of sulfide under microaerobic conditions at DO $<$ 0.1 mg/L in a batch reactor (Pokorna-Krayzelova et al., 2018). The model included biochemical oxidation of hydrogen sulfide to elemental sulfur by SOB, biochemical oxidation of elemental sulfur to sulfate by SOB, chemical oxidation of hydrogen sulfide to thiosulfate, thiosulfate disproportionation to elemental sulfur and sulfur dioxide, the decay of SOB and the hydrogen sulfide acid-base reaction. Under microaerobic conditions (DO below 0.1 mg/L), elemental sulfur was the major end-product of both biochemical and chemical sulfide oxidation. In the presence of bacteria, approximately 60% of the elemental sulfur was formed through biochemical oxidation of sulfide and 40% through the chemical pathway.

7. Design considerations and operations for micro-aeration

7.1. Point of air/oxygen injection

Micro-aeration can be applied in the headspace, the reactor bulk fluid, or a separate dedicated reactor. In a review of seven operating facilities in Central Europe, Jení ček et al. noted that all the reactors studied applied micro-aeration in sections of piping returning sludge to the anaerobic digesters (Jenicek et al., 2017). The concern noted with application to the headspace was based on the potential for biofilm growth fouling gas piping and/or equipment. The application of oxygen or air in the headspace or the sludge recirculation piping could achieve comparable hydrogen sulfide removal rates, with the latter resulting in an increase in overall oxygen demand, attributed to other reactions in the bulk liquid.

7.2. Quantity of air

For the application of micro-aeration in existing digesters, design centers on the provision of sufficient oxygen/air for biogas desulfurization. Full-scale data reported by Jení ček et al. and Kobayashi et al. were used in Fig. 2a (Plants B, K, Ko, L, M, N, and U are from Jeníček) and show that in full-scale AD, the mass of air supplied varied linearly with the mass of H_2S removed (Jenicek et al., 2017; Kobayashi et al., 2012). Given that oxygen constitutes 23% by weight of air, the ratio of oxygen consumed to H_2S removed is approximately 2.05 kg O_2/kg S^o removed, which is four times the theoretical value based on oxidation to sulfur. Ramos and Fdz-Polanco reported that 43%− 55% of the air injected into the headspace of pilot mesophilic anaerobic digesters was used for H_2S removal (Ramos and Fdz-Polanco, 2014). Diaz et al. supplemented 0.94 and 1.27 Nm^3 air/m^3 feed to the headspace of a 200 L mesophilic digester fed with sewage sludge, and reported that only 30–40% of the total oxygen supplied to the system served for H_2S oxidation, while the rest left the bioreactor with the produced biogas (Díaz et al., 2010) However, Giordano et al. who tested micro-aeration in three full-scale thermophilic digesters in series, each with working and headspace volumes of 4000 and 500 m^3, respectively at HRT of 44–71 days over 81 weeks observed that 13%− 59% of the consumed oxygen was not involved in the partial oxidation to sulfur (Giordano et al., 2019). Furthermore, based on the aforementioned study data, the ratio of air added to hydrogen sulfide removed at the lowest airflow was 1.85 kg O_2/kg S^o. It thus appears that while in pilot systems, 1.8–3.3 times the theoretical air requirement was required, in full-scale systems, about 4 times the theoretical air requirements are needed, potentially due to mixing and oxygen transfer as well as different surface area-to-volume ratios. Fig. 2b shows the linear correlation between the air-to-biogas flow ratio and the mass rate of sulfide removed. Fig. 2c shows the relationship between the ratio of air added to specific biogas flow per unit digester liquid flow, and the mass of H_2S removed based on the same full-scale data set discussed above and shown in Figs. 2a and 2b. The linear correlation is high and indicates that the air-to-specific biogas flow is roughly 23 times the sulfide mass removal rate. It should be noted that typically specific biogas production (daily biogas in m^3/d, divided by digester volume in m^3) in full-scale plants is about 0.3, as reported by Bodík et al. (2011) who monitored the largest 10 municipal wastewater treatment plants in Slovakia for over a year . This correlation, developed for a broad range of initial (500–5000 ppmv of H_2S) and effluent (10–1000 ppmv H_2S) concentrations, can be used in conjunction with Fig. 2a for the design of full-scale micro-aeration systems in mesophilic digesters. Giordano et al., using full-scale thermophilic digester data, determined that the air-to-biogas % mass ratio varies linearly with the residual % O_2 in the biogas, i.e., the air-to-biogas % mass ratio increases by five times the increase in residual O_2 in the biogas (Giordano et al., 2019). Considering the densities of biogas and air, this implies that air-to-biogas % volume ratio increases by 4.4 times the % increase in residual O_2 in the biogas, or conversely for every 1% increase in volumetric air-to-biogas, the oxygen percentage in the headspace increases by 0.22%. From the above correlations, knowing the biogas flow and H_2S concentration in a digester and the target H_2S, the mass rate of H_2S oxidized and theoretical oxygen required can be determined. Using the factors of 4 and 22.7 derived above, the applied oxygen mass rate and/or airflow rate can be determined, and thus the expected composition of the biogas can be estimated. In the subsequent

figures, capital letters indicate full-scale unnamed plants in Janicek et al., 2017.Fig. 2.

7.3. Liquid Surface Area-Headspace Volume Ratio

Unfortunately, there are few studies that examined the impact of liquid surface area-to-headspace volume ratio on biogas desulfurization. This parameter is significant, as the liquid surface area impacts oxygen transfer rates into headspace biofilms that serve as a reservoir for sulfide oxidizing bacteria. Mulbry et al. compared two 65-L (35 cm diameter, 60 cm height) agricultural anaerobic digesters with a working liquid volume of 6 L (59 L headspace) and liquid surface areas of 0.099 and 0.15 m^2 for biogas desulfurization using micro-aeration; the unit with 50% more surface area achieved three times the H_2S oxidation rate of the control but twice the oxidation rate when normalized to surface area. Maximum removal rates corresponded to 40–100 g S m^{-2} d^{-1} of liquid surface area at biogas retention times of only 30–40 min (Mulbry et al., 2017). Further evaluation into this relationship is needed to better quantity sulfide oxidation capacity for different digester configurations.

7.4. Biogas Residence Time

The importance of biogas residence time in gaseous hydrogen sulfide removal efficiency has been highlighted by Khoshnevisan et al., which compiled data from 11 studies and established that biogas residence times of five hours or more ensured around 90% removal efficiencies (Khoshnevisan et al., 2017). Fig. 3a shows that the removal of H_2S followed a first-order model, with a rate constant of 12.5 d^{-1}. While this data may indicate that the gaseous H_2S concentration is limiting biodegradation by SOB, the very strong linear correlation between removal rate and initial (background) H_2S concentration, depicted in Fig. 3b, indicates that the removal rate is independent of the H_2S concentration in the biogas with micro-aeration i.e., zero-order with respect to the biogas H_2S.

Biogas H_2S concentrations after micro-aeration in the 11 studies compiled by Khoshnevisan et al. ranged from 35 ppmv to 960 ppmv (average of 290 ppmv). Using the reported dimensionless Henry's constant of H = cG/cL = 0.5, with cG in ppmv and cL in moles/l (Krayzelova et al., 2015), the estimated liquid phase concentrations range from 2.4 to 65.8 mg/L. Thus, the concentrations of H_2S in the water vapor on the walls of the digester above the liquid surface, assuming equilibrium, with the gaseous H_2S, would also range from 2.4 to 65.8 mg/L. de Zwart et al. (1997b) reported a Monod half-saturation concentration (Ks,S^{2-}) for pure culture of Thiobacillus thioparus of 0.32 mgS^{2-}/L. Xu et al. (Xu et al., 2013) estimated Ks,S^{2-} for SOB grown from activated sludge at 11 mgS^{2-}/L while Fedorovich et al. estimated Ks,S^{2-} of 3.2 mgS^{2-}/L (Fedorovich et al., 2003). Notwithstanding the significant variability in Ks,S^{2-}, it is apparent that, on average basis, the dissolved H_2S was generally higher than the Monod half-saturation concentration and approached the zero-order rate. On the other hand, residual oxygen in the gas was around 1% (Krayzelova et al., 2015), and thus the dissolved oxygen in the headspace water vapor is about 0.25 mgO_2/L. Xu et al. reported a half-saturation concentration for dissolved oxygen for SOB of 3.2 mgO_2/L, and accordingly gaseous H_2S removal would be limited by oxygen rather than sulfides (Xu et al., 2013). This explains the first-order fit illustrated in Fig. 3a and the importance of sulfide removal as a function of gas residence time.

Based on the statistical correlations of full-scale data developed for the following design and operational conditions:

- Digester volumes: 300–30000 m^3
- Biogas flow rates: 500–9600 m^3/d
- H_2S concentrations in biogas: 500–5000 ppm
- Biogas residence time: 1.4–8 h

The following microaeration system design procedure, based on a safety factor (O_2 required /theoretical O_2 needed to oxidize H_2S to sulfides) of 4, is suggested:

1. H_2S removal efficiency is estimated using the first-order rate of 0.52 h^{-1} and the biogas residence time. The mass rate of H_2S removed (kg/d) can then be estimated.
2. Air flow mass rate = 8.94 * H_2S mass rate removed. This can be translated to an equivalent air flow.
3. Air flow (m^3/d) = 22.73 *H_2S removed (kg/d) *biogas flow (m^3/d)/ digester volume (m^3). Design air flow is the larger of air flows in steps 2 and 3.
4. Expected % oxygen in headspace oxygen concentration = 0.22 *volumetric percentage of airflow/biogas flow

8. Process instrumentation and control

8.1. Instrumentation

There are commercially available analyzers for measuring methane, carbon dioxide, oxygen, and hydrogen sulfide in digester biogas, but no direct online measurement of dissolved hydrogen sulfide in the liquid phase. Aside from biogas analyzers, full-scale anaerobic digesters are usually sparsely equipped with online instrumentation, particularly with regard to VFAs and sulfides (Wu et al., 2019). Monitoring VFAs is not only beneficial for monitoring general digester health but is also a parameter of importance with regard to SRB populations; SRBs may be able to utilize lactate and other fermentative products to produce VFAs (Moestedt et al., 2013). Consequently, populations of SRBs may still be high in low-sulfate feedstocks.

Lackner et al. (2020) studied the impact of SRB on thermophilic anaerobic digestion and observed that SRB degraded propionate and butyrate to acetate, and although acetate could be utilized by both SRBs and methanogens, acetate accumulation ensued and the acetoclastic methanogen genus Methanosarcina was inhibited by H_2S. Furthermore, SRBs have a higher affinity for H_2 than methanogens (Stams et al., 2005) and thus favor the growth of VFA-oxidizing, H_2-producing acetogens.

Despite the availability of various chromatography-based online measurements of VFAs (headspace gas chromatography, e.g.), widespread use is constrained by rapid clogging of the various membrane filters (Jimenez et al., 2015). Spectral techniques such as UV/visible spectroscopy (UV/vis), Mid InfraRed Spectroscopy (MIR), and Near Infra-Red Spectroscopy (NIRS) have been demonstrated as effective online monitoring tools. UV/vis has been demonstrated for the measurement of COD, TOC (Sarraguça et al., 2009), and VFAs (Wolf et al., 2013), while Krapf et al. (Krapf et al., 2013) demonstrated the use of NIR for VFAs measurement; however, commercialization of spectral techniques in AD is hampered by cost. Wu et al. reviewed the use of biosensors like microbial electrolysis cells (MEC) and microbial fuel cells (MFC) for constituent detection (Wu et al., 2019). Ranges of detection for VFAS (<12,000 mg/L) were satisfactory for AD, but calibrations were conducted with synthetic substrates. Recently, Lamb et al. presented an emerging technology based on colorimetric sensor arrays for VFA measurements that, if successful, could be completed within 2 min, and costs significantly less than other titrimetric, chromatographic, and spectroscopic methods. Furthermore, the technology could be applied for the measurement of ammonia at > 50 ppb and hydrogen sulfide at 50 ppb-50 ppm. (Lamb et al., 2019).

A useful indicator of the anaerobic conditions in the digester is the oxidation-reduction potential (ORP), which is a more sensitive parameter than dissolved oxygen. ORPs below − 300 mV are recommended to ensure no adverse impact on methanogens (Jimenez et al., 2015). Furthermore, ORP can be used as an indicator of sulfate reduction and biogas sulfide control in micro-aeration (Nghiem et al., 2014).

Given the lack of sufficient online measurement tools for the characteristics pertinent to micro-aeration, models could be used. However, the use of models to predict dissolved hydrogen sulfide is complicated.

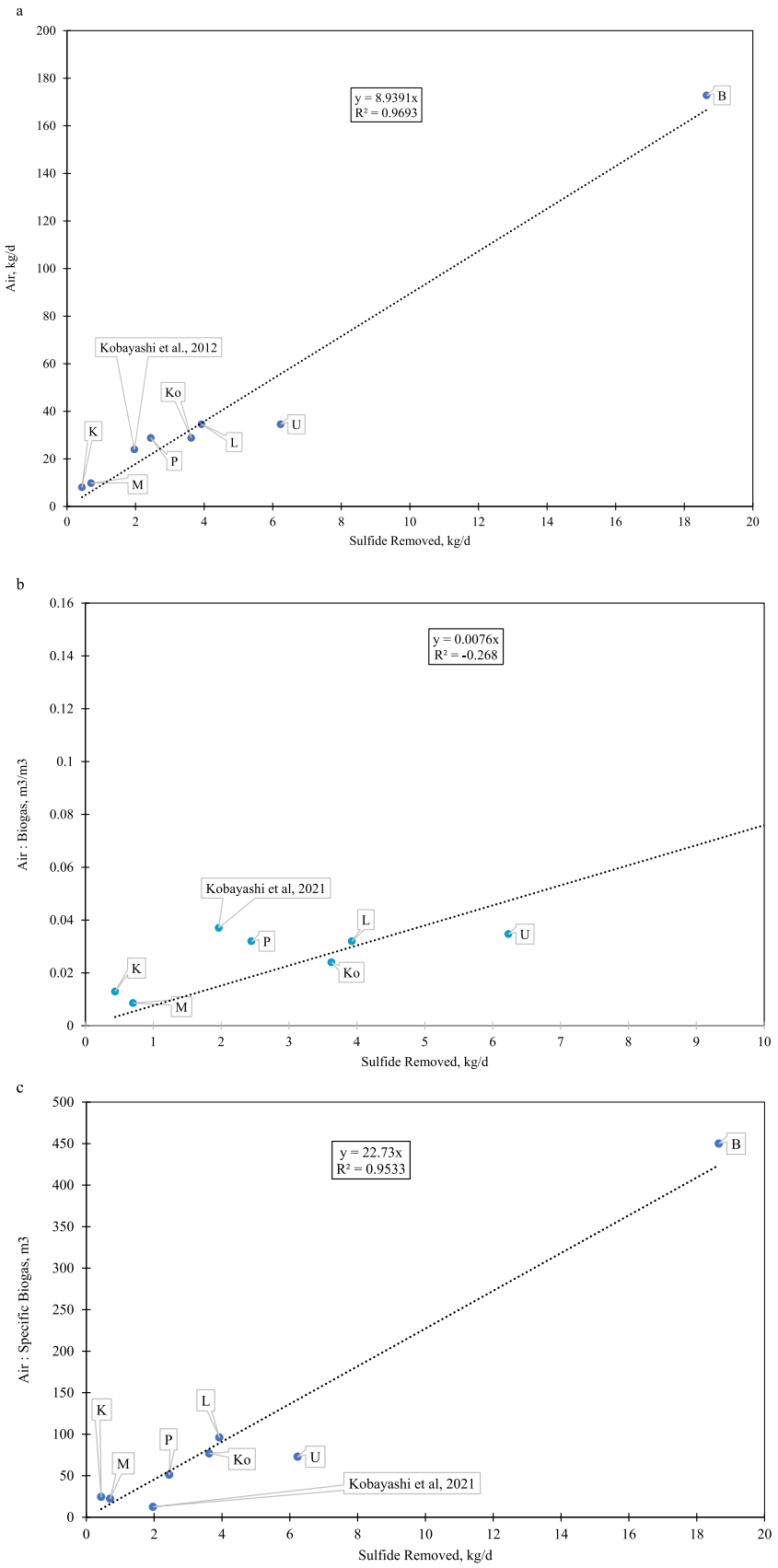


Fig. 2. a: Air Flow versus Sulfide Removed Fig. 2b: Air to Biogas Ratio versus Sulfide Removed Fig. 2c: Air to Specific Biogas Ratio versus Sulfide Removed.

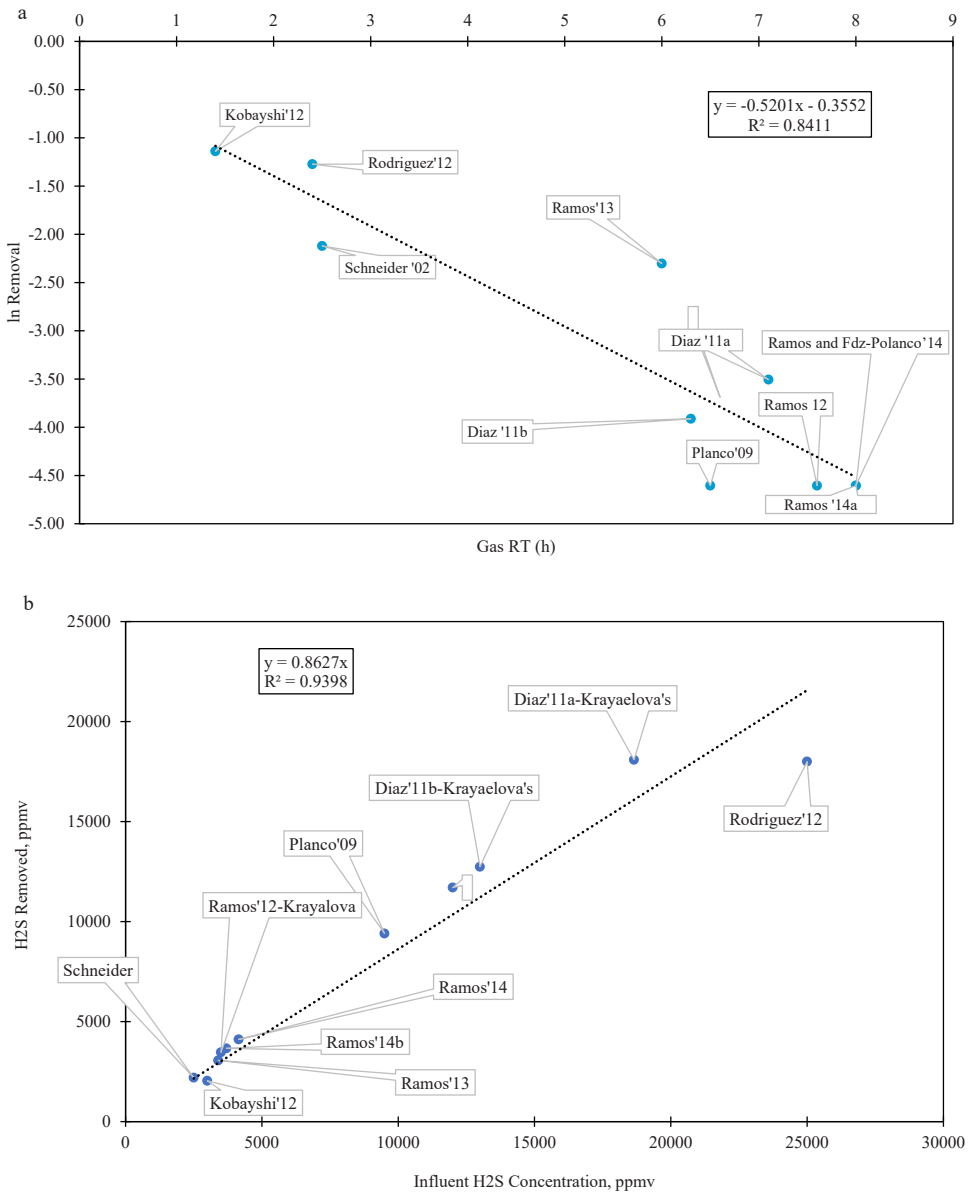


Fig. 3. a: First Order Kinetics of Removal versus Gas Residence Time. Fig. 3b: Zero Order Kinetics of Removal versus Influent Hydrogen Sulfide Concentration.

Models like ADM (Batstone et al., 2002) require input variables that are difficult to characterize experimentally, particularly when extension of ADM for sulfur speciation relies on insufficient knowledge of SOB in oxygen-limited environments and precipitation reactions involving sulfur.

8.2. Control

The adoption of developed control strategies in full-scale anaerobic digestion is hindered by the lack of reliable, accurate, and inexpensive online instrumentation (Gaida et al., 2017). In anaerobic digesters, the only manipulated variables are generally feed sludge flow and characteristics. In full-scale plants, where micro-aeration is used for sulfide control in digesters, the main manipulated variable is the air/oxygen flow since sludge flow and characteristics are dictated by upstream processes. Various control variables have been used in micro-aeration studies, predominantly at the lab and pilot-scale levels. Ramos and Fdz-Polanco used biogas flow and biogas H_2S concentration to regulate air flow in 200-L mesophilic anaerobic digesters treating primary and waste sludge at an HRT of 19 days; average sulfide removal efficiencies were 99%, and oxygen concentrations in the biogas of less than 0.08% v/v were reported at pure oxygen-to-biogas flow ratios of 0.35%– 0.5% (Ramos and Fdz-Polanco, 2014). The control of oxygen was based on the discrepancy between the actual H_2S concentration in the biogas and the target limit of 100 ppmv, with a controller sampling time of 30 min. It should be noted that although theoretically there is a risk of sustained oxygen overdosing if biogas flow is the control variable, as excess air/oxygen would increase the apparent biogas flow, given that the typical oxygen:biogas ratio in full-scale digesters is around 1% (Krayzelova et al., 2015), which is less than the variability of biogas flow as a result of intermittent feeding and decanting, the actual risk of overdose is significantly mitigated. Since the oxygen feed to the digester is controlled by specific feedstock composition and operation, digester-specific control points, such as using the biogas sulfide concentration for variable feedstock composition, should be employed. Microaeration control based on dissolved oxygen is impractical considering the very low ambient DO in highly reductive systems such as anaerobic digesters, and the accuracy and limitations of analytical methods such as the Winkler method, electro-chemical sensors, and optical sensors (Magdalena et al., 2022), and the response time of online

sensors which may lead to overshooting.

ORP under micro-aeration is 0 to − 300 mV; strict anaerobic conditions veer closer to − 400 mV. Various studies have used ORP as the means of process control with various results. ORP in the range of − 265 mV to − 230 mV (without micro-aeration) have been demonstrated to achieve > 98.5% H_2S removal efficiency from the treatment of sulfate-rich wastewaters in CSTRs (Khanal and Huang, 2003) and upflow anaerobic filters (Khanal and Huang, 2006). Nghiem et al. maintained an ORP of − 310 mV to − 290 mV in a micro-aeration study using a 50 L anaerobic digester and achieved approximately 99.5% reduction of H_2S (Nghiem et al., 2014). Table 4 summarizes studies that have used ORP as process control and the associated ranges and key observations.

Despite the feasibility of developing site-specific correlations between ORP and dissolved oxygen, the main drawback of an ORP-based control is that ORP is system-specific and thus extrapolation from one plant to the other is difficult. Moreover, most full-scale digesters are not equipped with ORP sensors and for those that are, ORP control can be unreliable due to solids accumulation on the sensor. Thus, experimental data without micro-aeration must be generated prior to establishing a target ORP control strategy for micro-aeration based on the differential ORP between micro-aerobic and background.

Finally, due to the lack of reliable sensors for liquid phase total dissolved sulfide, coupled with the fact that changes in liquid phase sulfide concentrations are usually 20%− 30% of the changes in gaseous hydrogen sulfide (Krayzelova et al., 2014), there are no studies that have attempted to control micro-aeration with dissolved sulfides. While most of the aforementioned process control studies have focused on avoiding oxygen overdosing through maintaining ORP and oxygen levels in the headspace, a recent study by Andreides et al. (2021) explored the optimization of H_2S removal by shaving off peaks from an 20-L anaerobic SBR treating cheese whey using stirring-based control strategies and reported that gradual increase in mixing speed from 15 to 35 rpm affected a 97% reduction at H_2S concentration of 5800 ppm.

9. Perspectives on future research

There is a definite paucity of information regarding several aspects of micro-aeration in digester biogas desulfurization. First, consistent kinetic data of SOB should be collected. Varying test conditions and data derived from studies operating at high DOs are not reflective of typical micro-aeration applications at very low DO. Additionally, using pure SOB cultures rather than mixed anaerobic digestates is not representative. This data would enable the development of simpler semi-empirical models, based on detailed full-scale data for various mixing regimes in digesters, as well as integrated mathematical models to reflect not only chemical and biochemical oxidation but also the impact of micro-aeration on SRB and SOB. Although this study did identify a few integrated model studies, they were all calibrated with data from synthetic wastewater studies, and to varying degrees did not reflect aerobic carbon oxidation, re-reduction of elemental sulfur back to sulfide, potential oxidation of sulfides to sulfates, methane production, and did not consider SRB inhibition and competition between SRB and methanogens. Data on the impact of digester surface area on the efficiency of micro-aeration desulfurization and full-scale data on the effect of biogas residence time are also lacking. Finally, from an analytical perspective, the development of more reliable, cost-effective, and rapid tests for dissolved sulfide will be crucial for process control. The impact of various mixing devices and air/oxygen injection strategies on the diurnal H_2S variability in full-scale digesters needs further investigation, as it impacts not only, oxygen and H_2S mass transfer rates, which in turn affect headspace concentrations, but also biogas residence time and desulfurization efficiency. Furthermore, considering the potential oxidation of hydrogen sulfide to sulfates instead of sulfur, detailed sulfur and COD balances from pilot and full-scale facilities would help quantify the various mechanisms involved.

Table 4
Summary of ORPs in Micro-aeration.

Reference Study	ORP Range (mV)	Notes
Yin et al. (2016)	-100/ − 200 mV	None
Khanal and Huang et al. (2003)	-230/ − 180 mV	Demonstrated the feasibility of ORP for Control of Sulfides in synthetic wastewaters at sulfate concentrations of 1000–5000 mg/L. ORP increases of 50 and 100 mV above strict anaerobic conditions were tested. The required increase in ORP was dependent on the influent sulfate concentration.
Liu et al. (2013)	-50/ − 350 mV	Optimized fermentation process for ethanol, butanol, propanediol
Donoso-Bravo et al. (2018)	< −494 mV	Bench reactor, avoiding any O_2 toxicity potential

10. Conclusions

Interest in anaerobic digestion as a renewable energy is increasing, but capital costs associated with biogas cleaning and upgrading can be an economic barrier to implementing AD projects. Microaeration is a promising technology for simple, economical, robust, and reliable hydrogen sulfide reduction that can improve the economics of gas systems and potentially provide process benefits in the digester itself. Additional data and research should be collected for full-scale projects to further inform, and fill knowledge gaps associated with design and control of microaeration systems. Development of more reliable dissolved sulfide measurement methods as well as more sensitive and responsive dissolved oxygen sensors can facilitate process control and avert oxygen overdosing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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